gfi / Israel

MeatCODE*

Open Flavor & Aroma Initiative

Promoting meaty flavor solutions via a collaborative effort and shared toolbox – for accelerating adoption of alternative meat products.

Daniel Dikovsky, Lior Teper
GFI IL SciTech

*CODE=Collaborative Open Development Ecosystem

My Why


Kobe
Black Angus
Holstein Female
REDEFINE MEAT


MPS
TD100-xr


COURAGE
PASSION
CURIOSITY

What Will Move the Adoption Needle?

The alternative meat performance envelope

True competition with animal meat requires balancing multiple drivers at once.

MOST IMPORTANT DRIVER

Taste parity

flavor • texture • juiciness

Price parity

cost • accessibility

Perceived health

nutrition • clean label

Convenience

prep • versatility

Environmental impact

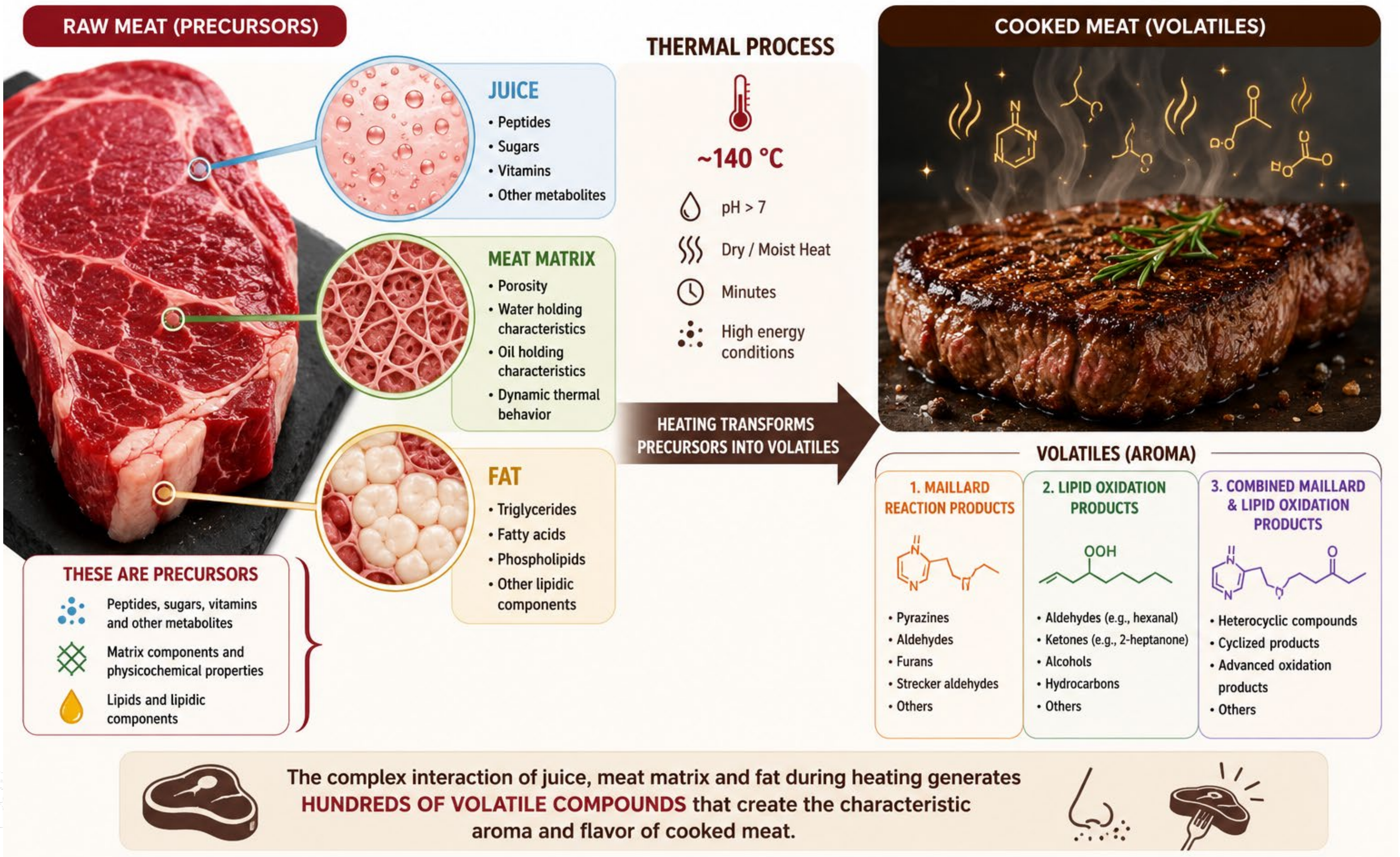
climate • resources

Critical enabling advantage for builders and mission-driven stakeholders.

Winning products must compete across the full envelope — not just on one attribute.

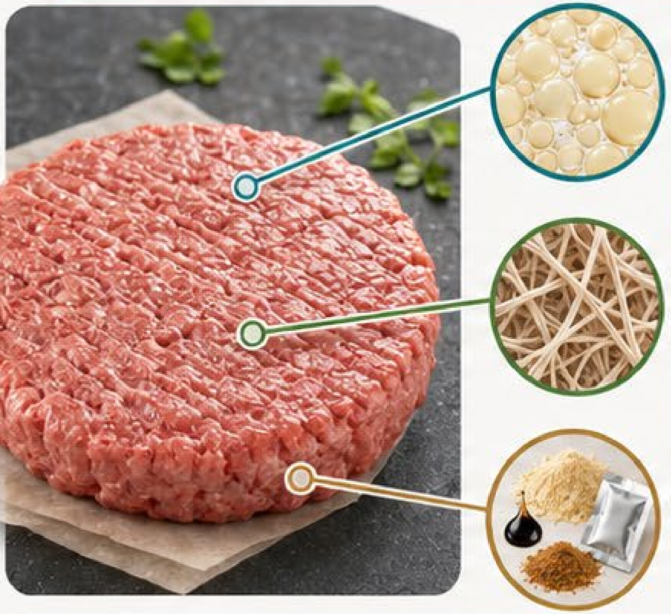
Consumer adoption is driven mainly by taste, price, health perception, and convenience, while environmental impact remains a critical enabling advantage.

HOW MEATY PROCESS FLAVOR WORKS IN ANIMAL MEAT



HOW MEATY FLAVOR WORKS IN MEAT ANALOGS

RAW MEAT ANALOG
(PRE-ADDED FLAVORS + PRECURSORS)



PLANT FAT
- Coconut oil
- Canola oil
- Sunflower oil
- Palm oil
- Other high-melting plant fats

PROTEIN MATRIX
- TVP / HME
- Texturized proteins
- Fibrous structure
- Water holding
- Oil holding

FLAVOR AGENTS
- Flavor masking additives
- Yeast extracts
- Hydrolyzed proteins
- Nutritional yeast derivatives
- Added reaction / artificial flavors

KEY DIFFERENCE
- A substantial share of meaty flavor is already added in the raw product
- Flavor houses pre-generate many meaty notes before formulation
- The burger also still contains some heat-reactive precursors

COOKING / THERMAL TRANSFORMATION
- Pan / grill heating
- Browning
- Moisture loss
- Aroma release
- Additional Maillard-type reactions

HEATING RELEASES ADDED FLAVORS AND CREATES SOME NEW VOLATILES

COOKED MEAT ANALOG
(AROMA / VOLATILES)

VOLATILES (AROMA)

1. ADDED FLAVOR-DERIVED AROMA
- Pre-made reaction flavors
- Yeast / hydrolysate flavor notes
- Released during heating

2. PROCESS-FORMED AROMA
- Maillard-type products
- Some lipid-derived products
- Formed from available peptides, sugars, vitamins

3. COMBINED / FINAL AROMA PROFILE
- Added + newly formed volatiles
- Cooked meaty character
- Residual plant notes may remain

In meat analogs, much of the meaty aroma is **pre-generated and added before cooking,** while cooking **mainly releases** these flavor systems and **creates a smaller share** of new process-flavor volatiles.

Current solutions are not enough because the field is fragmented

Meaty flavor is not one missing molecule. It is an integrated process-flavor system that must work at product level.

1 Partial system mimicry

Most solutions still rely heavily on added flavors rather than recreating the cooking-driven flavor system.

2 Scattered knowledge

Relevant science is spread across Maillard chemistry, lipids, fermentation, sensory, matrix physics and meat science.

3 IP + paywall barriers

Much applied know-how is locked inside companies, patents, flavor houses or inaccessible literature.

4 No shared language

Methods, terminology, benchmarks and target molecules are not sufficiently harmonized for collaboration.

5 High entry barrier

New players repeat avoidable mistakes because the complete integrated picture is hard to access.

MeatCODE addresses the coordination failure: it makes the system visible, searchable, and actionable.

2026 – GFI IL to Launch an Initiative to Accelerate Cooperative Progress Towards Meaty Flavor Solutions

Proposed Intervention

1. **Build an open platform** that turns scattered knowledge into shared resources, tools, and practical guidance — leveraging AI where helpful.
2. **Increase awareness** of the challenge and the opportunity across the ecosystem, catalyzing coordinated action.
3. **Map white spaces** and direct research toward the most impactful technical targets.
4. **Use GFI's positioning, network, and mission** to create global leverage and collaboration.
5. **Deliver a practical toolbox** for day-to-day R&D work, while capturing feedback and learnings from users.
6. **Activate focused research efforts** to address critical bottlenecks rapidly and efficiently.

Better meaty flavor is within reach — if the field works in a more coordinated way.

The Proposal – MeatCODE Initiative

One platform for coordinated, focused effort with real impact

MeatCODE Initiative

AI-POWERED PLATFORM FOR MEATY FLAVOR R&D

Knowledge Hub
Curated science, data and insights in one place

Expertise Directory
Find experts, capabilities and organizations fast

Molecular Database & Analytics
Integrated molecular, sensory and process data with powerful analytics

R&D Toolbox: protocols, methods & benchmarks
Standardized, tested and benchmarkable resources

Simulation & Modeling Tools
Predict mechanisms, volatiles and sensory outcomes

Collaborative Co-creation
Workspaces, projects and shared IP frameworks

Continuously Updated via AI Agents
AI agents monitor, learn, and surface what matters most

Designed for many user types and usage scenarios

Researchers · Flavorists · Product Developers · Companies · Academia

GFI nucleates — the community joins as value is created.

A tool that should become a natural part of meaty flavor R&D work

MeatCODE turns fragmented knowledge and effort into a connected, intelligent system that accelerates discovery and drives real-world impact.

Who will use MeatCODE, and how we will know it works

Primary users

Academic labs
Flavor chemistry, fermentation, lipids, proteomics, analytics, meat science

Flavor + ingredient companies
Yeast extracts, hydrolysates, lipids, masking, delivery systems

Alt-meat startups + food companies
Product teams seeking practical routes to better meaty flavor

GFI teams + funders
White-space mapping, project design, funding prioritization

Impact indicators

Research engagement and publications
active users • repeat users • expert contributors

R&D activation
new collaborations • white-space projects • grant/funding leads

Tool adoption
protocols used • datasets cited • toolbox downloads

Field outputs
publications • patents • prototypes • product-development programs influenced

Success means not only traffic. It means better decisions, fewer repeated mistakes, and more focused R&D activity.

What success looks like by stage

2026

MVP validated

Expert users test a source-backed hub; priority white spaces and first toolkits are released.

2027

Recognized reference hub

MeatCODE becomes a natural starting point for meaty flavor R&D teams and collaborators.

2028

Prototype-level impact

Hub-supported projects generate integrated process-flavor prototypes and shared datasets.

2029

Traceable field influence

First pre-commercial or commercial programs show influence from MeatCODE-enabled knowledge, tools or collaborations.

2026 is not the final product launch. It is the validation year for a credible, useful, scalable GFI IL + GFI Labs flagship platform.

Key risks and mitigation plan

Risk	Mitigation	Control
Poor engagement / low usage	Build with anchor users; advisory board and demo-driven onboarding; measure repeat use, not only sign-ups.	**User testing**
Weak usability	Define 3–4 core user journeys; run MVP feedback loops; prioritize clarity over feature breadth.	**UX sprints**
AI reliability / hallucination	Use curated source base, citations, confidence tags, expert review and clear “unknown” behavior.	**QA protocol**
Scope too broad	Start with meaty process flavor MVP: literature, experts, molecules/pathways, methods and white-space map.	**Scope gates**
IP / paywall limits	Focus on public-domain knowledge, metadata, patents, non-confidential summaries and pre-competitive gaps.	**IP guardrails**
Insufficient R&D impact	Tie hub outputs to Labs experiments, benchmark protocols and partner projects; track project creation.	**Activation KPIs**
Resource overload	Lean team, AI tooling, selected subcontractors, partner contributions and fundraising options for scale-up.	**Monthly review**

The biggest risk is trying to build the complete vision too early. The mitigation is a narrow MVP with rapid expert validation.

Lean execution: narrow MVP, fast validation

1 Core team

Daniel + Lior as product owners, domain leads and platform builders.

2 AI + subcontractors

Claude Code and other AI tools; lightweight technical/UX support where needed.

3 Project discipline

Asana board, weekly rhythm, decision log, risk register and milestone reviews.

Execution principles

MVP first

Build the first functional package that creates real value; avoid premature platform complexity.

Build with users

Interview, demo, test and adjust with selected academic, industry and GFI users.

Use the network

Leverage GFI affiliates, advisory board candidates, Labs data and external partners.

Prepare for scale

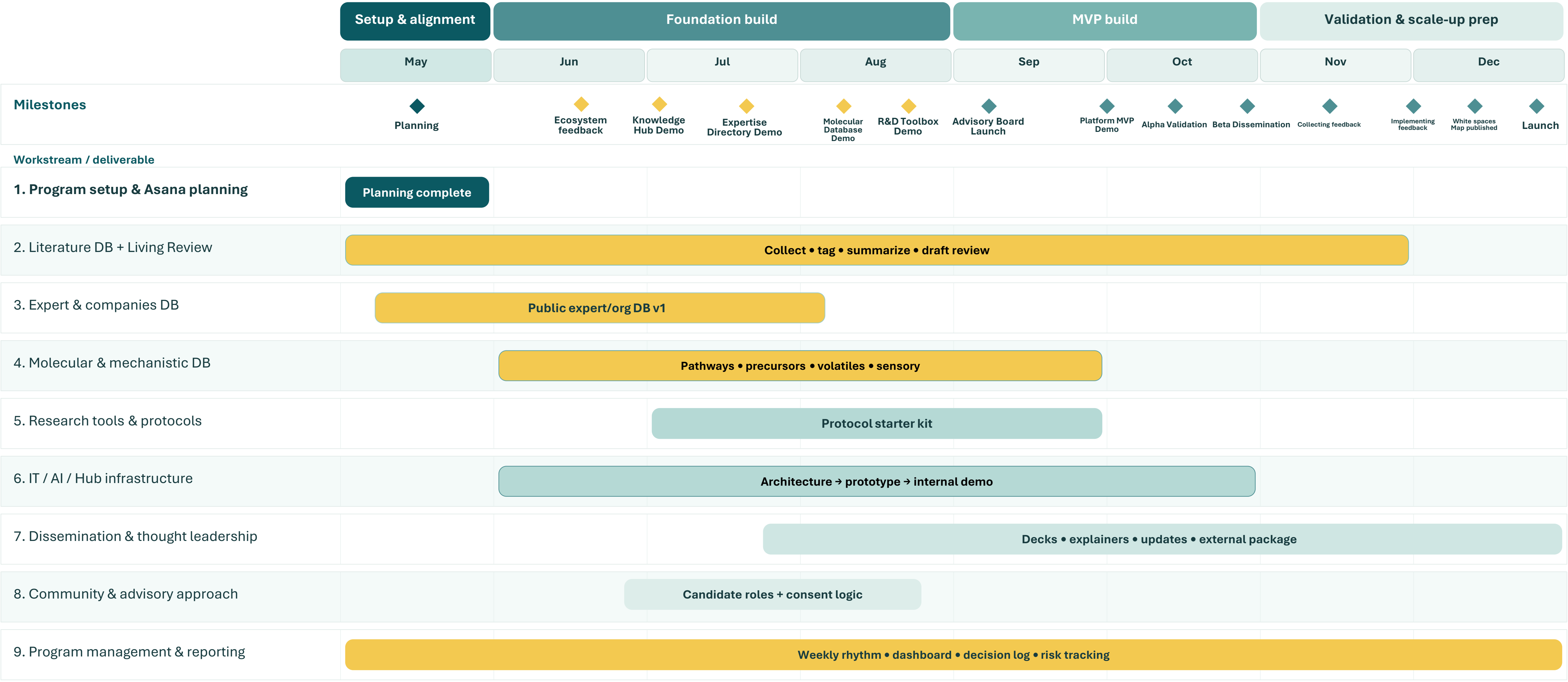
Identify one strong academic partner, one strong industry partner and funding routes for 2027.

Operating promise: credible progress without pretending we already have a full product.

MeatCODE – Open Flavor & Aroma Initiative

Execution timeline | May–Dec 2026

MVP focus: build the first credible seed of a meaty process flavor knowledge hub, with structured data assets, research tools, and validation-ready outputs.



completed · in progress / near-term · upcoming milestone

May status: scoping + Asana planning completed; Literature DB, Expert DB, Molecular DB, and Research Tools are now active.

gfi / Israel

Good Food Institute

How Labs by GFI Israel contributes to MeatCODE

Labs serves as the experimental backbone that helps MeatCODE generate, validate, and sharpen actionable knowledge.

Labs by GFI Israel

1. Generates anchor data
Produces high-quality experimental data on flavor, aroma, matrix and processing behavior.

2. Validates hypotheses
Tests ideas emerging from literature, patents, and expert insights.

3. Explores white spaces
Runs focused experiments in high-priority gaps identified by MeatCODE.

4. Benchmarks methods
Establishes practical protocols, reference systems, and comparison frameworks.

5. Creates real-world use cases
Turns platform questions into concrete research programs with practical relevance.

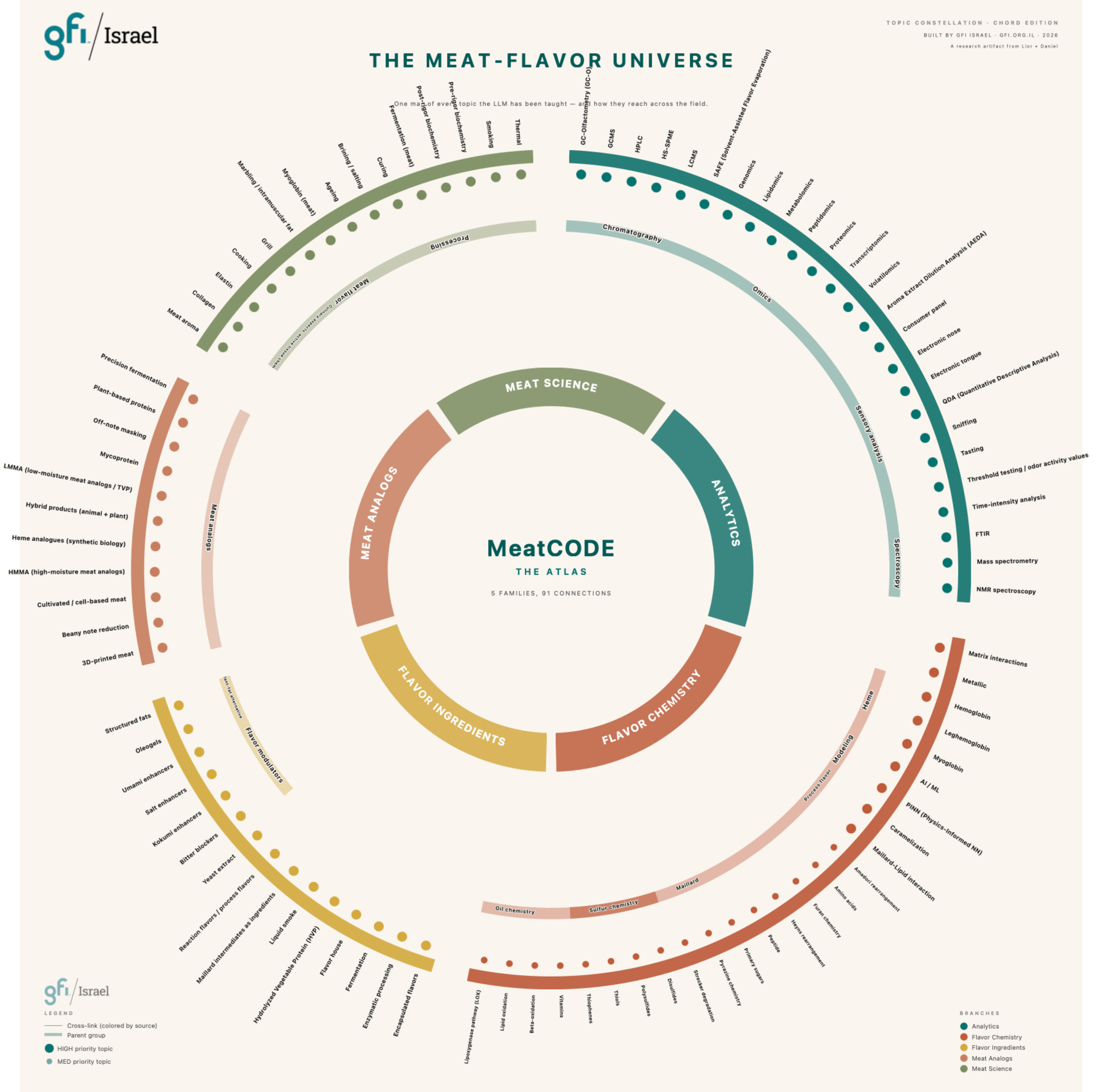
Labs projects that can feed MeatCODE

- MFL
- Generic Platform
- Hydrolysates
- FAT
- Blends
- Other targeted flavor studies

Takeaway: *Labs is not just a stakeholder — it is a primary engine for validating and enriching MeatCODE.*

The Knowledge Hub Construction

- Mapping the field – topics, search criteria, sources mining and refinement
- DB construction and population
- Preliminary analysis and curation
- Search expansion, tagging and structural analysis
- UI and Chat bot engineering



MeatCODE platform demo

A working seed of the knowledge hub, not yet a polished product. The purpose is to test value and direction.

Live demo by Lior Teper

Please watch for usefulness, traceability, user experience and missing features.

Literature DB

source-backed answers

Expert directory

who knows what

Molecular layer

precursors • volatiles • sensory

Research workflows

queries • summaries • white spaces

After the demo: we will collect management feedback on scope, usability, positioning and support needed for the 2026 MVP.

MeatCODE achievements so far

A functioning seed platform is already in place, with core assets, tools, and early ecosystem pull.

Execution foundation

- Team functional
- Asana plan in place
- AI tools operational

Platform readiness

- Database established
- App platform functional
- Chatbot initialized

Knowledge & ecosystem

- Literature DB: 1,000+ papers and books
- Expertise directory and Molecular DB initiated
- Strong external interest: Wageningen, Masha Niv, FSI

Next proof point: Live platform demo and focused feedback on usability, value, and next-build priorities.

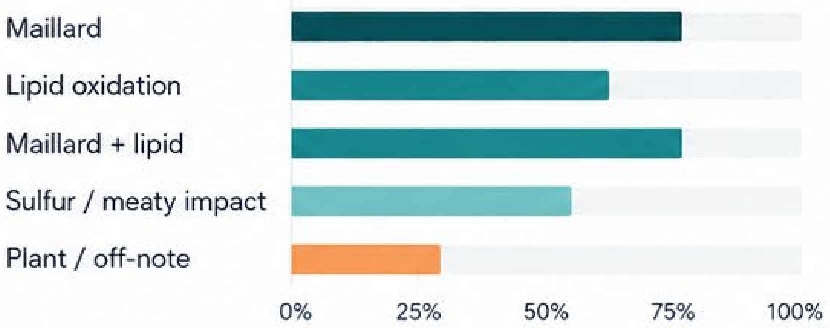
Concept demo: MeatCODE Volatile Atlas

How GC-MS data becomes interpretable meaty-flavor insight

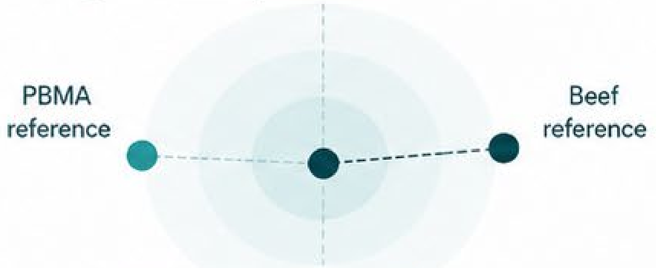
1. Upload GC-MS table → 2. Annotate volatiles → 3. Build pathway fingerprint → 4. Compare to references → 5. Generate flavor insights

Prototype sample: PBMA-07 | Target: Grilled beef burger

A. Pathway profile



B. Target similarity



Closer to beef, but still missing some meaty drivers

C. Top drivers

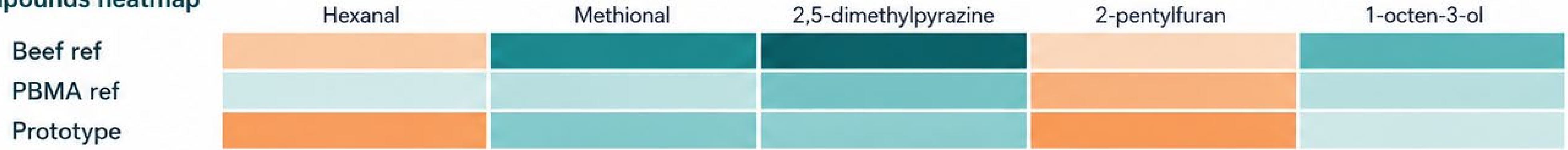
Toward target

- Pyrazines
- Methional
- Roasted furans

Away from target

- Hexanal excess
- 2-pentylfuran
- Earthy/beany notes

D. Key compounds heatmap



Interpretation layer
Adds meat-flavor meaning to raw compound lists

Benchmarking
Frames each sample against meat and PBMA references

Decision support
Highlights missing, excessive, and promising aroma drivers

Proposed collaboration: MeatCODE GC-MS Flavor Visualization Module

Turning GC-MS volatile data into actionable meaty-flavor insight

Goal

Create a joint MeatCODE module that visualizes GC-MS volatile data through the lens of meat flavor chemistry, helping researchers interpret what moves a sample toward or away from convincing meaty aroma.

Why it matters

GC-MS datasets are rich but often hard to translate into flavor insight. The module adds interpretation by linking volatiles to pathways such as Maillard, lipid oxidation, and Maillard + lipid interactions, and by comparing samples to meat and PBMA reference profiles.

What the module should contain

- Curated volatile annotation layer
- Pathway-based dashboards
- Reference comparison to beef / chicken / pork / PBMA datasets
- Gap-to-target analysis and top flavor drivers
- Exportable summary for experimental decision-making

Proposed first collaboration phase

- Build a proof-of-concept with public + selected internal datasets
- Curate a first dictionary of key meat-related volatiles
- Develop a visual dashboard and interpretation rules
- Use it as a basis for future joint publications and wider MeatCODE deployment

Potential value for Wageningen + MeatCODE

Scientific interpretation layer for GC-MS | Shared benchmark framework for meaty flavor | Open demonstrator for alt-meat R&D

How Can You Help Move MeatCODE Forward?

Help turn an early working seed into a useful, credible, and shared ecosystem resource.

SCI

Scientific partnership

Contribute feedback, validation logic, references, datasets, and technical thinking that strengthen the scientific backbone.

LABS

Experimental support

Use Labs by GFI IL and related projects to generate anchor data, test hypotheses, and explore high-value white spaces.

NET

Network and dissemination

Help connect the initiative to relevant experts, collaborators, communications channels, and potential external partners.

LEAD

Leadership and enablement

Support prioritization, sponsorship, funding discussions, and the practical steps needed to keep the initiative moving.

The immediate ask: endorse MeatCODE as a GFI IL + Labs by GFI IL 2026 initiative, nominate contributors, and help shape the first practical modules.

If we do this well, MeatCODE can become shared infrastructure for meaty flavor knowledge, benchmarking, and collaborative progress.